

FRAMES *THAT* SPEAK

*An augmented-reality museum treasure hunt where movement,
close looking, and in-place interpretation are the learning mechanism.*

Julian Bauer · Shruti Khetade · Arindam Tripathi

Motivation & Background

Why this matters — and why AR is not automatic.



The Attention Problem

Studies show most visitors spend only 15–30 seconds in front of an individual artwork. More time looking → more meaningful experiences.



Information is Not the Bottleneck

Wall labels and audio guides deliver facts but pull attention away from the artwork itself. Visitors need support for knowing what to notice.



AR as a Candidate

AR could keep interpretation attached to the artwork — anchoring context, directing the eye. But it's only justified if it does more than deliver the same content through a novel interface.

State of the Art & Gaps

What existing systems do — and what they miss.

Museum AR Overlays

A 2022 meta-analytic review of 51 museum learning studies found AR produced positive effects on achievement. Yet most systems superimpose supplementary materials — stopping short of structured learning.

Engagement ≠ Learning

One art-museum study found AR increased visitors' liking of artworks more, while a book guide produced stronger learning gains. Engagement and learning do not automatically follow each other.

Living-Portrait Precedent

Snap's 2026 Jonathan Yeo exhibition and Louvre collaboration showed AR-powered portraits that respond in real time are viable — but neither is designed around a learning structure.

Gap: No system combines AR overlays + living portrait moments inside a theme-based learning path.

Proposed Approach

Three workstreams, integrated into one experience.



AR Recognition & Anchor Pipeline

Unity + AR Foundation: image tracking per artwork, spatial anchoring of overlays, content delivery from recognition to in-scene interaction.

Unity · AR Foundation · ARKit / ARCore



Interaction Engine & Themed Hunt Logic

Clue progression, theme-based branching, prompt triggering, state management across artworks, and UI flow for treasure-hunt completion.

C# · State Machines · UI Toolkit



Generative Living Portrait Pipeline

Image-to-video workflow turning static paintings into animated portrait moments, integrated into AR via contextual triggers.

Image-to-video models · Python · Asset pipeline

Challenges

Recognition



How does the system know which artwork is in front of the user?

Anchoring



How do overlays stay visually attached to a real painting as the user moves?

Integration challenges for both with Meta SDK



While Pre-existing drag and drop frameworks for both tasks exist on mobile devices, like AR Kit and Vuforia, it is currently not possible to integrate them easily onto the quest platform

Recognition and Anchoring approaches

QR Marker Recognition approach

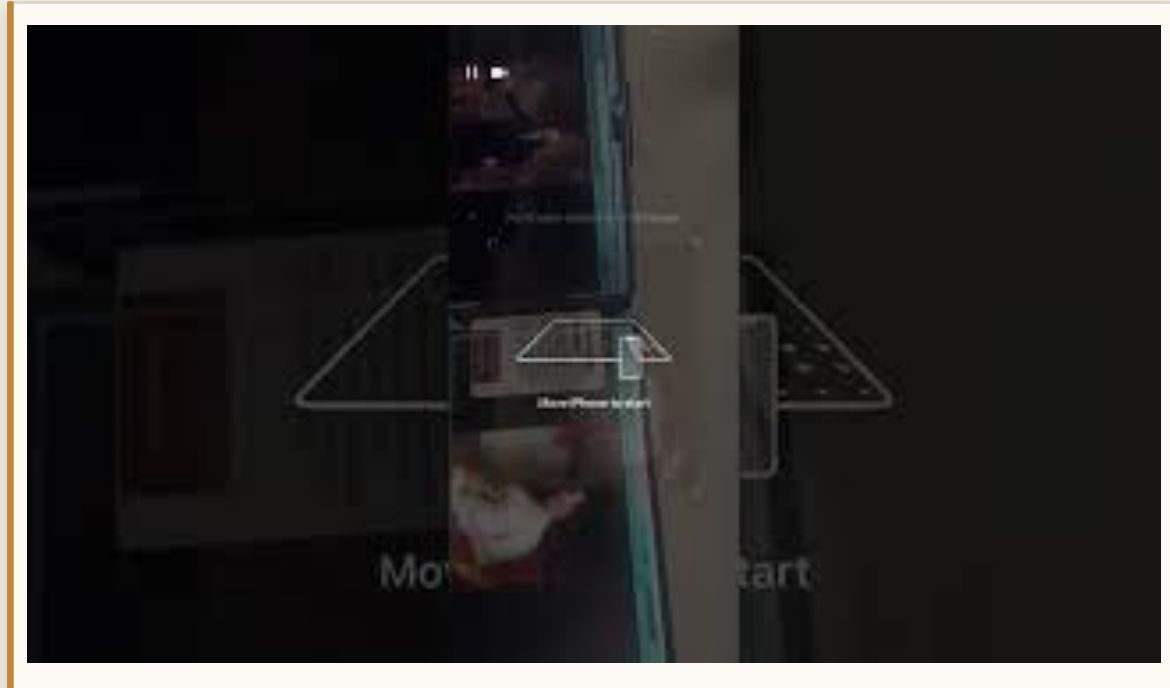
We explored the QR anchoring approach where Meta Quest detects a physical QR code as a trackable anchor and read its payload like `portrait:demo_01`. That payload is mapped to a specific video/portrait prefab, which is spawned on the portrait.

We did not use QR anchoring because every portrait would need a visible QR code. Also, QR markers must be printed large enough and clearly visible for reliable tracking, making the approach less practical for natural museum displays.

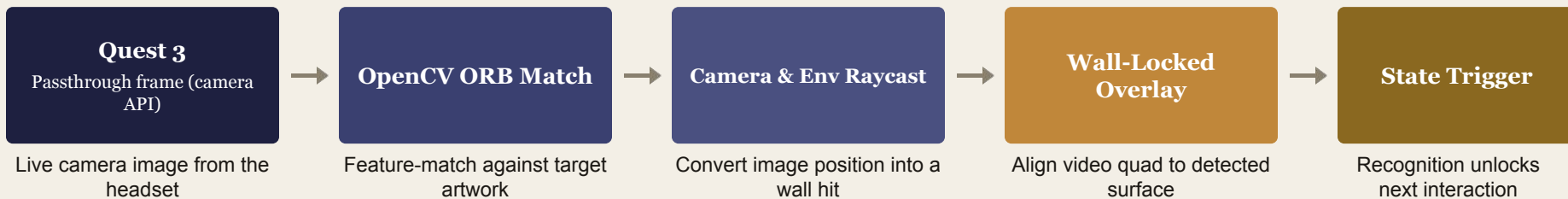
ARToolKit and Scenery based mobile approach

An alternate mobile AR approach was also explored using ARToolKit and a Scenery-based workflow, where image recognition and overlay placement were tested on a phone-first AR stack before moving fully into the Quest pipeline. This helped validate the basic interaction pattern. However, due to issues with integration with the quest platform, we had to shift to a different approach.

Recognition and Anchoring Phone Demo



Recognition & Anchoring Implementation



Target Matching

One active artwork target is loaded at a time to ensure high performance.

Spatial Placement

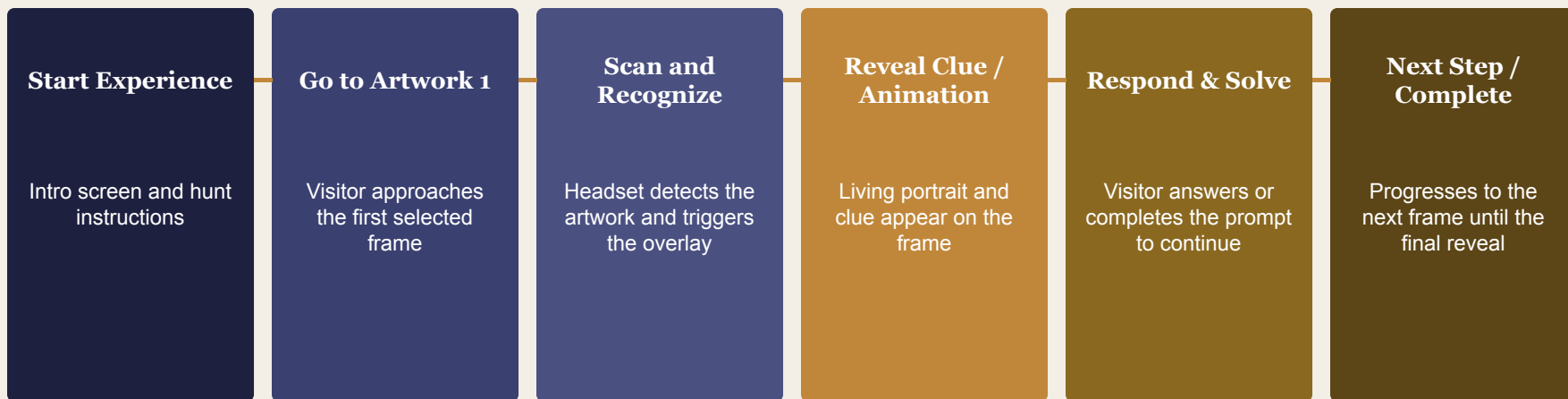
Environment raycast anchors content directly to the physical wall surface.

Runtime Stability

Overlay freezes and detection pauses after lock to maintain stable alignment.

The system recognizes the artwork in the headset camera feed, converts the detection into a physical wall hit, then locks the living-portrait overlay to that surface.

The user flow



The experience moves the visitor through a linear sequence of scanning, revealing, and responding, with each artwork unlocking the next step in the hunt.

What Was Accomplished

IMPLEMENTED BY US

- | | |
|--|-------------------------|
| ✓ Quest 3 AR museum scene | <i>Arindam + Julian</i> |
| ✓ Artwork recognition + anchor stability | <i>Arindam</i> |
| ✓ Game Logic | <i>Julian</i> |
| ✓ Clue / overlay UI flow | <i>Shruti</i> |
| ✓ Living-portrait generation workflow | <i>Julian + Shruti</i> |
| ✓ Generative output → AR integration | <i>Shruti + Arindam</i> |
| ✓ Staged gallery + demo walkthrough | <i>All</i> |

BORROWED COMPONENTS

- | | |
|---|--|
|  Unity Engine | Core development platform |
|  Meta SDK Building blocks | Camera Rig, Passthrough, Environment Raycast Manager |
|  XR Interaction Components | Spatial interaction primitives |
|  OpenCV | Image Recognition and anchoring |
|  Image-to-video model (NeroAI) | Generative living-portrait backbone |
|  Public artwork images (National Portrait Gallery open access Repository) | Inputs to the portrait pipeline |

Results & Demo



Quest 3 AR museum scene

Staged gallery with printed art reproductions and spatial overlays.



One complete hunt walkthrough

End-to-end: recognition → clue → overlay
→ living portrait → completion.



Generative portrait integration

Static paintings animated via image-to-video pipeline, triggered on recognition.



Live Demo: Staged Gallery Walkthrough



Live Website: Frames that Speak

Future Work

Robust Artwork Recognition



Replace manual selection with automated recognition across diverse lighting and viewing angles.

Multiple Gallery Themes



Support symbolism, patronage, composition, and historical context across the same collection.

Large User Study



Run a structured study measuring recall, attention duration, and learning outcomes across themed hunts.

Glasses-Based Deployment



Port to Snap Spectacles or similar hardware for hands-free, in-gallery use at scale.

THANK *YOU!*